

Reionization driven by the few: the ionizing budget of galaxies at $z=5-10$ from JWST/NIRSpec

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ABSTRACT

The sources responsible for the Epoch of Reionization (EoR), the last major phase transition of the Universe, remain highly elusive. While JWST has begun to illuminate the properties of early galaxies, the direct detection of their ionizing photons is virtually impossible due to the neutral inter-galactic medium (IGM) at $z \gtrsim 6$. A direct estimate of the escape fraction of ionizing photons f_{esc} is thus not possible. However, the escape fraction is encoded in spectral features that trace a low opacity to ionizing photons, namely the reduction of nebular emission with increasing f_{esc} . Here, we exploit the large archive of JWST/NIRSpec spectra at $5 < z_{\text{spec}} < 10$ from the DAWN JWST Archive to provide new constraints on the ionizing emissivity of EoR galaxies and on the timeline of reionization. Through detailed spectral fitting of 1428 galaxies with a picket-fence model for the escape of ionizing photons, we derive posteriors of f_{esc} together with the effective ionizing output. This approach yields the escaping ionizing output of each galaxy while bypassing the usual decomposition into the ionizing photon production efficiency (ξ_{ion}) and f_{esc} . Combining these measurements with UV luminosity functions (LFs), we derive the hydrogen ionizing LFs, resulting in an observationally-derived ionizing photon budget at $5 < z < 15$, integrated down to $M_{\text{UV}} = -13$. Under standard assumptions, our budget yields an evolution of the IGM neutral fraction consistent with independent probes, with reionization ending at $z \sim 5.8$. Given the evolving shape of the ionizing LF, bright sources ($M_{\text{UV}} \sim -20$) dominate the cosmic ionizing emissivity $\dot{n}_{\text{ion}}$ at lower redshift ($z = 5 - 6$), whereas at $z > 6$ faint galaxies ($M_{\text{UV}} > -18$) contribute roughly equally. Remarkably, a mere $\sim 20\%$ of sources, those with $f_{\text{esc}} > 10\%$, produce $\sim 87\%$ of the ionizing photons. Overall, our results support a picture in which a few strongly leaking galaxies drive most of reionization.

Key words. galaxies: high-redshift – cosmology: dark ages, reionization, first stars – early Universe

1. Introduction

The epoch of reionization (EoR) is the last major phase transition of the Universe, when the hydrogen in the intergalactic medium (IGM) transitioned from completely neutral to completely ionized (Becker et al. 2015; Dayal & Ferrara 2018; Robertson 2022; Fan et al. 2023; Ellis 2025; Stark et al. 2026, for some recent reviews), finally allowing Lyman Continuum (LyC) photons ($\lambda_{\text{rest}} < 912 \text{ \AA}$) to travel through it mostly unimpeded. This process started with the formation of the first luminous objects (Barkana & Loeb 2001) and was likely driven by star-forming galaxies (e.g. Bouwens et al. 2015; Robertson et al. 2015; Dayal et al. 2025). However, there is still debate about whether faint (Atek et al. 2024) or bright (Naidu et al. 2020) galaxies dominate this transition. Active Galactic Nuclei (AGNs) have also been suggested as important contributors to this process (Giallongo et al. 2015; Madau & Haardt 2015; Grazian et al. 2024; Madau et al. 2024), although studies on their luminosity function in the EoR suggest that their contribution is likely minor, compared to that of galaxies (Mitra et al. 2018; Hassan et al. 2018; Kulkarni et al. 2019; Jiang et al. 2022; Matsuoka et al. 2023).

In this context, the process of reionization is expected to be very patchy (e.g., D’Aloisio et al. 2015; Jamieson et al. 2025), with bubbles of ionized gas forming and growing around the objects that leak the most ionizing photons. This is seen directly in observations of the Gunn-Peterson trough, which show field-to-field variations towards the end of the EoR (Bosman et al. 2018; Eilers et al. 2018; Yang et al. 2020; Bosman et al. 2022; Meyer et al. 2025). Although inhomogeneous, the global evolution of reionization can be summarized with the volume filling fraction of ionized hydrogen $Q_{\text{HII}}(z)$ and its counterpart, the fraction of neutral hydrogen x_{HI} . There have been significant efforts to constrain x_{HI} and its redshift evolution, both from single bright objects that can probe the Gunn-Peterson trough, such as quasars, and from the redshift evolution of galaxy populations. Constraints on x_{HI} have been placed from observations of the Lyman- α and Lyman- β absorption in the spectra of quasars (Eilers et al. 2018; Kulkarni et al. 2019; Yang et al. 2020; Bosman et al. 2022; Qin et al. 2025) and galaxies (Meyer et al. 2025), the clustering of Lyman- α emitters (Sobacchi & Mesinger 2015; Ouchi et al. 2018), the Lyman- α equivalent width and luminosity function evolution (Ouchi et al. 2010; Konno et al. 2014, 2018; Zheng et al. 2017; Inoue et al. 2018; Mason et al. 2018, 2019a; Hoag et al. 2019; Whitler et al. 2020; Goto et al. 2021; Morales et al. 2021; Bolan et al. 2022;

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Ning et al. 2022; Bruton et al. 2023; Morishita et al. 2023; Nakane et al. 2024; Tang et al. 2024; Kageura et al. 2025) as well as the damping wing of either Lyman-break galaxies (Hsiao et al. 2024; Umeda et al. 2024; Mason et al. 2026) or quasars (Schroeder et al. 2013; Davies et al. 2018; Wang et al. 2020; Greig et al. 2019, 2024). All these different estimates place the end of the reionization process at $z \sim 5.3 - 6$.

In its most basic form, the redshift evolution of the fraction of neutral hydrogen is calculated as the balance between the recombination time scale and the rate of ionization (Madau et al. 1999). The number of ionizing photons that reach the IGM per unit time and unit volume, the cosmic ionizing emissivity, is described as

$$\dot{n}_{\text{ion}} = \rho_{\text{UV}} f_{\text{esc}} \xi_{\text{ion}}. \quad (1)$$

Here ρ_{UV} is the UV luminosity density (in units $\text{erg s}^{-1} \text{Hz}^{-1} \text{Mpc}^{-3}$), derived from the UV luminosity function, f_{esc} is the average escape fraction of ionizing photons of the general galaxy population and ξ_{ion} is the average ionizing photon production efficiency (in units Hz erg^{-1}), measuring the amount of ionizing photons created per $1500 \text{ \AA}$ luminosity density.

Since the launch of the James Webb Space Telescope (JWST, Gardner et al. 2023) we have gained an unprecedented view of the rest-frame optical properties of galaxies in the EoR, both thanks to the Near-Infrared Camera (NIR-Cam, Rieke et al. 2023) and the Near-Infrared Spectrograph (NIRSpec Jakobsen et al. 2022). In particular, many studies have focused on separately measuring the three parameters that make up our understanding of the ionizing photon output of the Universe, ρ_{UV} (Bouwens et al. 2023; Donnan et al. 2024; Harikane et al. 2025; Whitler et al. 2025; Atek et al. 2026; Chemerynska et al. 2026; Kreilgaard et al. 2026; Weibel et al. 2026), ξ_{ion} (Simmonds et al. 2024a,b; Llerena et al. 2025; Pahl et al. 2026), and f_{esc} (Witten et al. 2023; Mascia et al. 2024b; Saxena et al. 2024; Giovinazzo et al. 2026; Stoffers et al. 2026; Jecmen et al. 2026). However, considering each component individually does not account for possible correlations between the parameters, which can have significant implications on the estimated ionizing budget (Muñoz et al. 2024; Bosman & Davies 2024). In particular f_{esc} is an exceedingly challenging quantity to measure as effectively all ionizing photons are absorbed by the IGM in the EoR, making direct measurements impossible. A complete analysis where all the parameters are modeled together, self-consistently, is therefore necessary to fully understand the reionization process.

Giovinazzo et al. (2026) estimated f_{esc} for a sample of sources with NIRSpec/PRISM spectroscopy in the EoR, from Spectral Energy Distribution (SED) fitting. In this work, we will expand on this analysis to estimate the rate of ionizing photon production ($\dot{N}_{\text{ion,int}}$) for the same sample, again using SED-modelling. As a result of the f_{esc} prescription already implemented by Giovinazzo et al. (2026), we can infer posterior distributions for the ionizing photon emission rate ($\dot{N}_{\text{ion,esc}}$). Similar efforts in estimating the ionizing emissivity from SED fitting have also been undertaken by Simmonds et al. (2024b), with a mass complete sample, although without accounting for f_{esc} . By estimating the ionizing photon emission rate directly through integration of the ionizing part of the spectral SED fits, we avoid having to disentangle the separate contributions of the production efficiency and the escape fraction, along with their associated uncertainties, to $\dot{N}_{\text{ion}}$. This yields a self-consistent estimate of the rate at which ionizing photons reach the IGM from a given

galaxy, a quantity necessary to constrain the reionization process.

In this paper, we will determine the evolution of the ionizing emissivity function of the general galaxy population as well as the cosmic ionizing emissivity. We will then calculate the redshift evolution of the neutral hydrogen fraction with cosmic time, as well as the contribution of strong leakers, here defined as $f_{\text{esc}} > 10\%$. We will test which population of galaxies contributes most to the ionizing emissivity, whether the few highly leaking galaxies or the majority of weakly leaking sources. We will also determine whether any galaxy population, i.e. at faint, intermediate or bright luminosities, is a dominant contributor of ionizing photons to the IGM.

In Sec. 2 we introduce the sources included in our sample, present the method that we used to calculate $\dot{N}_{\text{ion}}$ and compare it to the method typically used in the literature. In Sec. 3 we present the ionizing properties of the sources in our sample and their redshift and magnitude evolution. In Sec. 4 we present the ionizing emissivity function, quantifying the number of sources in a unit volume with a given rate of ionizing photon output. In Sec. 5 we present the evolution of the fraction of neutral hydrogen. Finally, in Sec. 6 and Sec. 7 we present our caveats and conclusions.

Throughout this work, we assume a flat Λ CDM cosmology with $H_0 = 70 \text{ km s}^{-1} \text{Mpc}^{-1}$, $\Omega_m = 0.3$, and $\Omega_\Lambda = 0.7$. Magnitudes are given in the AB system (Oke & Gunn 1983).

2. Ionizing output calculation

2.1. Spectroscopic data

For this study, we used the data set previously presented in Giovinazzo et al. (2026). This consists of 1428 sources with JWST NIRSpec/PRISM observations at $5 < z_{\text{spec}} < 10$ taken from the DAWN JWST Archive (DJA¹) (Heintz et al. 2025; Brammer & Valentino 2025), an online repository containing spectroscopic data from public JWST programs, all uniformly extracted and reduced with the same pipeline based on grizli² and MSAexp³. Further details on the data reduction and processing can be found in de Graaff et al. (2025); Heintz et al. (2025); Valentino et al. (2025); Pollock et al. (2025). The database query was performed on the 7th of February 2025 and includes all sources in the redshift range of interest with grade=3 (i.e., with robust redshift), $M_{\text{UV}} < -18$ and excludes known Little Red Dots (LRDs) as reported by Kocevski et al. (2025). This selection includes spectra from RUBIES (GO 4233, de Graaff et al. 2025) JADES (GTO-1180, 1181, 1210, 1286, 1287; GO-3215 Eisenstein et al. 2023) GTO WIDE (GTO-1211, 1212, 1213, 1214, 1215; Maseda et al. 2024), CEERS (ERS-1345; Finkelstein et al. 2023), UNCOVER (GO-2561, Bezanson et al. 2024), CAPERS (GO-6368; Donnan et al. 2025), GO-1433 (Hsiao et al. 2024), GO-2198 (Barrufet et al. 2025), GO-2565 (Nanayakkara et al. 2025), DD 2750 (Arrabal Haro et al. 2023), GO-3073 (Castellano et al. 2024), DD-2756 (Mascia et al. 2024b), DD-6541 (DeCoursey et al. 2025), DD-6585 (PI: Coulter) and GO-4106 (PI: Nelson).

2.2. Spectral SED fitting

For each source we perform spectral SED fitting to the NIR-Spec/prism data to recover the physical properties, as described

¹ <https://dawn-cph.github.io/dja/>

² <https://github.com/gbrammer/grizli>

³ <https://github.com/gbrammer/msaexp>

in Giovinazzo et al. (2026). These fits were performed with bagpipes (Carnall et al. 2018, 2019; Lange 2023) including a picket-fence model (Heckman et al. 2001; Zackrisson et al. 2013) to model the escape fractions. This formalism for f_{esc} was first introduced as a custom version of bagpipes in Giovinazzo et al. (2026), but has since been included in the standard release, from version 1.3.3.

We modeled the star formation history with the continuity prior (Leja et al. 2019), a non-parametric prior, allowing for greater flexibility than parametric models. We modeled the star formation rate (SFR) in seven bins –with 5, 10, 50, 100, 200, 400 and 800 Myr– unless the age of the Universe at the source’s redshift was less than 800 Myr. In that case, the last bin ended at the age of the Universe. With the continuity prior, bagpipes fits for the ΔSFR between adjacent bins, which therefore adds a number of free parameters equal to the number of bins minus one. We used a Student-t prior with $\Delta\text{SFR} \in [-3, 3]$ and adopted $\nu = 2$ and $\sigma = 0.3$ following Leja et al. (2019). For f_{esc} we opted for a logarithmic prior with range $0.001 < f_{\text{esc}} < 1$. This prior was chosen since observations (Kreilgaard et al. 2024) as well as simulations (Choustikov et al. 2024; Yeh et al. 2023) suggest that most sources have little-to-no leakage. This was confirmed in Giovinazzo et al. (2026). We used the BPASS v2.2.1 stellar-population models (Stanway & Eldridge 2018) with the default broken-power-law initial mass function (IMF), with slopes of $\alpha_1 = -1.30$ for the $0.1 - 0.5M_{\odot}$ mass range and $\alpha_2 = -2.35$ for the $0.5 - 300M_{\odot}$ mass range, based on Kroupa et al. (1993). Metallicity was modeled with a logarithmic prior, in the range $0.01 < Z/Z_{\odot} < 0.5$. The stellar and nebular attenuation were modeled with the Calzetti dust law (Calzetti et al. 2000), with a linear prior, allowing for a maximum $A_V = 0.5$ mag as we were mainly interested in blue sources. We note that for our picket-fence model dust is not applied to the free channels. For more details on the SED fits we refer the reader to Giovinazzo et al. (2026).

The SED fits are used to calculate the ionizing photon production for each source, as discussed in Sec. 2.4. This is then compared to the more standard approach, presented in Sec. 2.3, of measuring the ionizing photon production from the $H\alpha$ line.

2.3. $\dot{N}_{\text{ion}}$ estimation from $H\alpha$

In the standard approach used in the literature, the production rate of ionizing photons in the EoR is estimated from the luminosity of the $H\alpha$ line (e.g. Bouwens et al. 2016; Llerena et al. 2025; Pahl et al. 2026). As a recombination line, the $H\alpha$ luminosity directly depends on the amount of reprocessed ionizing photons and therefore effectively counts the number of ionizing photons that do not escape. For all the sources in our sample with $H\alpha$ coverage in the PRISM wavelength range, the $H\alpha$ fluxes are extracted following Covelo-Paz et al. (2026), i.e. by fitting a Gaussian directly to the spectrum. The luminosities are then corrected for dust attenuation following the Calzetti et al. (2000) dust model with attenuation A_V derived from SED fitting, as $L_{H\alpha, \text{int}} = L_{H\alpha, \text{obs}} \times 10^{0.4A_{H\alpha}}$.

The relation between the intrinsic Balmer line luminosity and the total Lyman Continuum luminosity was derived from stellar models by Leitherer & Heckman (1995). For $H\alpha$ it can be expressed as

$$L(H\alpha)[\text{erg s}^{-1}] = 1.36 \times 10^{-12} N(H^0)[\text{s}^{-1}] \quad (2)$$

where $N(H^0)$ is the production rate of ionizing photons assuming $f_{\text{esc}} = 0$, and $L(H\alpha)$ is the intrinsic $H\alpha$ luminosity, which is thus

dust-corrected. This effectively assumes that all the ionizing photons are reprocessed in the $H\alpha$ line and none escape. Hence, the ionizing photon production that we find from this method must be corrected for the escape fraction. As seen in Giovinazzo et al. (2026), most sources have very low escape fraction, making this a good approximation. There is, however, a small fraction of sources with very high escape fraction, as the f_{esc} distribution approximates an exponential with a mean of 10%. The ionizing photon production rate is then given by

$$\dot{N}_{\text{ion, int}} = \frac{N(H^0)}{1 - f_{\text{esc}}}. \quad (3)$$

For the escape fraction measurements we use those already presented in Giovinazzo et al. (2026), which are based on spectral SED fitting returning the full posterior distributions.

2.4. $\dot{N}_{\text{ion}}$ estimation from SEDs

Alternatively, the ionizing photon production rate can also be calculated as the integral of the spectrum, L_{ν} , below the Lyman limit ($\lambda < 912 \text{ \AA}$):

$$\dot{N}_{\text{ion}} = \int_{\nu_{912}}^{\infty} L_{\nu}(h\nu)^{-1} d\nu, \quad (4)$$

as is done for simulated galaxies (e.g. Naidu et al. 2020; Seeyave et al. 2023). We can apply this to calculate both the intrinsic ionizing photon flux, when integrating the intrinsic SED, and the escaping ionizing photon flux, when integrating the observed SED without IGM attenuation. In contrast to what is done with $H\alpha$, this method simply counts the photons that escape the galaxy.

Using the full posterior distribution of physical parameters for each source in the sample, we compute the posterior of both the escaping ($\dot{N}_{\text{ion, esc}}$) and the intrinsic ionizing photon flux ($\dot{N}_{\text{ion, int}}$) for each galaxy. Their respective uncertainties are also computed from the posterior distribution. In a first step, we create rest-frame SEDs, to remove the effect of IGM transmission, which still include both the dust and escape fractions to compute $\dot{N}_{\text{ion, esc}}$. Then, in a second step, we compute the intrinsic ionizing photon production of the stellar population ($\dot{N}_{\text{ion, int}}$) by removing dust obscuration, the effects of IGM transmission and setting $f_{\text{esc}} = 1$, to create an intrinsic SED. This results in a posterior distribution of both $\dot{N}_{\text{ion, esc}}$ and $\dot{N}_{\text{ion, int}}$ for each galaxy.

2.5. Comparison of ionizing fluxes

The two methods described before are complementary in how they count the amount of ionizing photons, as the $H\alpha$ method counts the rate of photons that are reprocessed into nebular continuum, while the SED integration directly counts the rate at which ionizing photons escape. Because of this, the only way to consistently compare these two methods is by using $\dot{N}_{\text{ion, int}}$, the intrinsic rate of ionizing photons production.

In Fig. 1 we compare the intrinsic ionizing photon production rate as estimated from the two methods. We use this comparison to ensure that our method is consistent with that primarily used in the literature (e.g., Bouwens et al. 2016; Llerena et al. 2025; Pahl et al. 2026). We find the $H\alpha$ estimates to be consistently lower, with a median difference between the estimates of $\Delta = 0.05$ dex, with only few sources lying significantly above or below the 1:1 line. This small discrepancy is likely due to the assumptions in the $N(H^0)$ conversion to $L(H\alpha)$

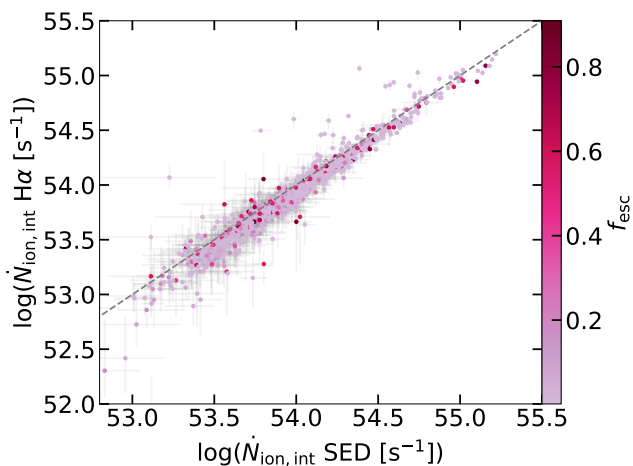


Fig. 1: Comparison between intrinsic ionizing photon production rates estimated with H α and from the SED fit, color coded by f_{esc} . The grey dashed line shows the 1:1 relation. We find a broad agreement, with small offset between the methods ($\Delta = 0.05$ dex), as well as some scatter, with only a few sources lying significantly off the 1:1 line. This agreement with a method widely used in literature (e.g. Bouwens et al. 2016; Llerena et al. 2025; Pahl et al. 2026) validates our SED fitting estimates.

of Leitherer & Heckman (1995) that do not completely match those of our bagpipes models. Under the assumption of case B recombination, different gas temperatures and densities between the bagpipes and the Leitherer & Heckman (1995) models could be the source of this discrepancy. Moreover, the two models differ in their IMF. While the high mass slope is the same ($\alpha = -2.35$), the mass range of Leitherer & Heckman (1995) only extends to 100 $M_{\odot}$, unlike the IMF used in BPASS that extends to 300 $M_{\odot}$. Young and massive stars are those expected to produce the most ionizing photons, and it has been shown by Schaerer et al. (2025) that Very Massive Stars (VMS; $M > 100M_{\odot}$) contribute significantly to the Lyman continuum emission of young stellar populations. In general, the stellar population synthesis code in Leitherer & Heckman (1995) is different than that of BPASS, especially in their lack of binaries, which also produce more ionizing photons than single stars. These factors may explain why $\dot{N}_{\text{ion,int}}$ as calculated from H α is underestimated when compared to that calculated from our SED fits. This suggests that estimates done with Eq. 2 may not be fully comparable to those that assume a different stellar population synthesis and spectral synthesis codes.

In Fig. 1 we also find some scatter with $\dot{N}_{\text{ion,int}}$ from H α occasionally being overestimated as compared to estimates from the SED. Visual inspection of these outliers shows that most have a broad H α line, hinting at the possible presence of AGNs or LRDs in these sources. We do not use an AGN template in our fits and therefore find a poor fit to the H α line for these sources.

Overall, we find good agreement between the two methods, indicating a good agreement between our SED fitting results, where f_{esc} is properly accounted for on a source-by-source basis and the method previously used in literature. This validates our model, which we will use for the remainder of the analysis to have access to the ionizing photon production rate also for sources where H α is not covered. In this work, we will mostly use $\dot{N}_{\text{ion,esc}}$ as we are interested in the rate at which ionizing photons reach the IGM.

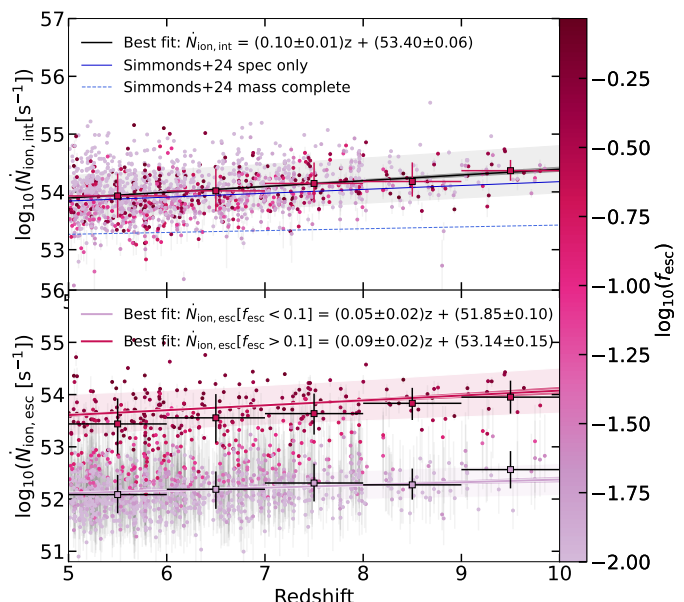


Fig. 2: **Top panel:** Redshift evolution of $\dot{N}_{\text{ion,int}}$. The black line shows the best fit for the redshift evolution, with the intrinsic scatter in the gray shaded area. The squares show the binned means. This evolution is consistent with the results from the spectroscopic only dataset from Simmonds et al. (2024b), shown as the solid blue line, but higher than the one from a mass complete sample, shown in the dashed blue line, also from Simmonds et al. (2024b). **Bottom panel:** Redshift evolution of $\dot{N}_{\text{ion,esc}}$, color coded by $\log(f_{\text{esc}})$. The pink line with the pink shaded area represents the fit to the $f_{\text{esc}} < 0.1$ sources and the red line with the red shaded area represents the same for sources with $f_{\text{esc}} > 0.1$. The two fits sit about 1.5 dex apart, with the high f_{esc} fit being much closer to the intrinsic $\dot{N}_{\text{ion,int}}$, as expected. The squares show the binned means, with the same color coding as for the linear fits.

3. Ionizing properties

3.1. Redshift evolution

Simmonds et al. (2024b) have shown that $\dot{N}_{\text{ion}}$ evolves with redshift when considering a spectroscopic-only sample, with higher ionizing emissivity at higher redshift. However, they have also demonstrated that this redshift evolution disappears when using a mass complete sample. This suggests a bias of the spectroscopic sample toward star forming galaxies rather than a real change in the ionizing emissivity with redshift for the global galaxy population. An important caveat of this previous analysis is the lack of a prescription for f_{esc} and therefore the assumption of $f_{\text{esc}} = 0$ for the calculation of $\dot{N}_{\text{ion}}$. We investigate the possibility of a redshift evolution of $\dot{N}_{\text{ion}}$, both intrinsic and escaped, in our sample, which includes a prescription for f_{esc} .

We find that the intrinsic ionizing photon production rate $\dot{N}_{\text{ion,int}}$ shows a shallow evolution with redshift, as seen in the top panel of Fig. 2. This trend may be due to a selection bias, as discussed by Simmonds et al. (2024b). It may also be due to a redshift evolution of M_{UV} or f_{esc} which would translate to a redshift evolution in $\dot{N}_{\text{ion}}$, the latter of which would have been missed in previous studies. In Giovanazzo et al. (2026), we do in fact see a shallow trend between redshift and f_{esc} for our sources, as also shown in Ferrara et al. (2025).

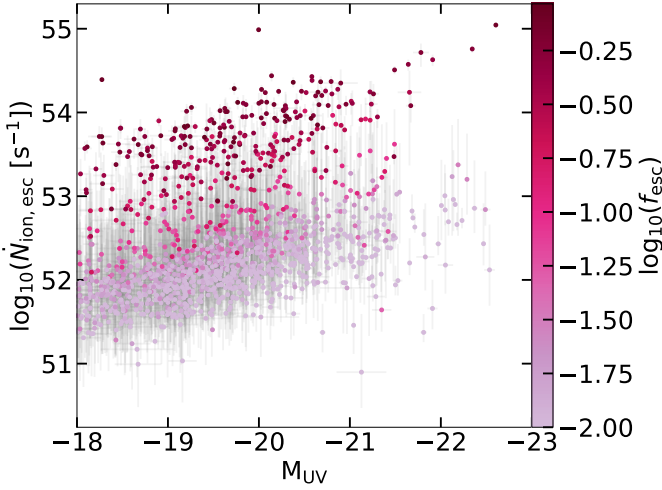


Fig. 3: Distribution of $\dot{N}_{\text{ion,esc}}$ as a function of UV magnitude color coded by f_{esc} . At the same magnitude, sources with increased f_{esc} also show higher $\dot{N}_{\text{ion,esc}}$, indicating that these are the sources that emit most ionizing photons into the IGM.

In the bottom panel of Fig. 2, we show the rate of escape of ionizing photons $\dot{N}_{\text{ion,esc}}$ against redshift. Here we also color-code the points by $\log_{10}(f_{\text{esc}})$ and see that the data seems to be split in two populations, one clustered around $\log(\dot{N}_{\text{ion,esc}}/\text{s}^{-1}) \sim 52$, with consistently low f_{esc} and another one with $f_{\text{esc}} \gtrsim 10\%$ which spans the range $53 < \log(\dot{N}_{\text{ion,esc}}/\text{s}^{-1}) < 55$. It should be noted that the points in the lower cloud, with low f_{esc} , may be more accurately interpreted as upper limits on $\dot{N}_{\text{ion,esc}}$ as the lowest possible value in the fit is $f_{\text{esc}} = 10^{-3}$, therefore not allowing for a true $f_{\text{esc}} = 0$. Moreover, as shown in Giovinazzo et al. (2026), with our spectral SED fitting method it is challenging to confidently constrain escape fractions below $f_{\text{esc}} \sim 20\%$ even for high signal to noise values, with f_{esc} usually being underestimated in this case. These uncertainties are properly captured by the posterior distributions, making the uncertainties on $\dot{N}_{\text{ion,esc}}$ for sources with low f_{esc} significant. The distribution may therefore not truly be bimodal, as it initially appears.

We additionally explore the redshift evolution of $\dot{N}_{\text{ion,esc}}$, as previously performed for the intrinsic value. We choose to use two linear fits to distinguish the population of strong-leakers ($f_{\text{esc}} > 10\%$) from that of weak-leakers ($f_{\text{esc}} < 10\%$). We choose to split the sample at $f_{\text{esc}} = 10\%$ as this is the mean f_{esc} of the entire sample, as shown in Giovinazzo et al. (2026). While both populations show a shallow redshift evolution, it is stronger for the high f_{esc} population, although the evolution is consistent within the uncertainties. This redshift evolution is closer to that of the intrinsic production rate, with a slope of 0.09 ± 0.02 , as most ionizing photons that are created are also emitted in the case of high f_{esc} . Instead, the low f_{esc} population shows a shallower distribution, with a slope of 0.05 ± 0.02 , about 1.5 dex below the strong leakers. As mentioned for $\dot{N}_{\text{ion,intr}}$ this redshift evolution could be due to a selection bias, or to a redshift evolution of M_{UV} or f_{esc} (e.g. Mascia et al. 2024a; Cain et al. 2025a; Ferrara et al. 2025). This would need to be studied further with mass-complete samples in the future. As described below, we will ignore any redshift evolution for the further analysis.

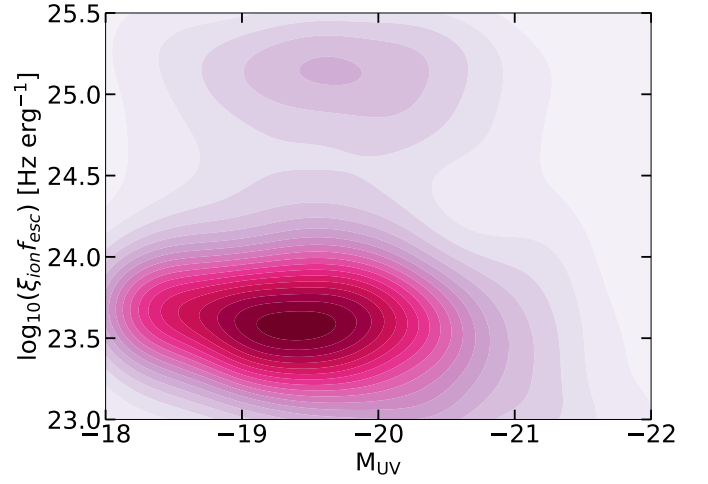


Fig. 4: Kernel density estimation of $\log(\xi_{\text{ion}}f_{\text{esc}})$ as a function of M_{UV} . The distribution shows negligible evolution as a function of M_{UV} . Moreover, it also shows two clouds. The main one, with the majority of our data, includes all the weak leakers, corresponding to the pink points in Fig. 3. The secondary cloud, above the main one, includes the sources with high f_{esc} , so the strong leakers.

3.2. UV magnitude dependence

In Fig. 3 we investigate the relation between $\dot{N}_{\text{ion,esc}}$ and UV magnitude. The ionizing photon production rate is proportional, to first order, to the rate of non-ionizing UV photon production. Hence, we expect $\dot{N}_{\text{ion,int}}$ to be proportional to the intrinsic UV luminosity. Indeed, it is immediately visible that $\dot{N}_{\text{ion,esc}}$ increases towards brighter M_{UV} . This allows us to investigate the impact of the escape mechanism on this relation. We color code the sample by f_{esc} , highlighting the same behavior already seen in Fig. 2. All of the low f_{esc} sources are clustered around $\log(\dot{N}_{\text{ion,esc}}/\text{s}^{-1}) \sim 52$, while the $f_{\text{esc}} \gtrsim 10\%$ sources span the range $53 \lesssim \log(\dot{N}_{\text{ion,esc}}/\text{s}^{-1}) \lesssim 55$, on average about 1.5 dex above the cloud of low f_{esc} sources. As mentioned above, however, the $\dot{N}_{\text{ion}}$ values are artificially limited in our fitting because of the limit on f_{esc} .

Overall, from Fig. 3 it remains clear that galaxies with the highest escape fractions also emit the most ionizing photons into the IGM, in absolute number. Indeed, above $\log(\dot{N}_{\text{ion,esc}}/\text{s}^{-1}) > 53$ only 8% of sources have an $f_{\text{esc}} < 10\%$. While faint galaxies with $f_{\text{esc}} \sim 50\%$ may emit as many ionizing photons as bright galaxies with only $f_{\text{esc}} \sim 10\%$, this result shows that f_{esc} is the key quantity to determine the sources of reionization, as it is the main factor controlling which sources leak a significant amount of ionizing photons. Indeed, even for very UV-bright sources that intrinsically produce a lot of ionizing photons, a very low escape fraction means that a negligible amount of them reach the IGM, even compared to fainter sources with an increased f_{esc} .

4. Ionizing emissivity function

We can remove the direct M_{UV} dependence by dividing $\dot{N}_{\text{ion,esc}}$ by the intrinsic UV luminosity of each galaxy, leaving us with Fig. 4, where we show $\log(\xi_{\text{ion}}f_{\text{esc}})$ against M_{UV} . Here we do no longer see the M_{UV} dependence, but we still see the difference between the strong and weak leakers.

In order to determine which sources contribute the most to reionization we must understand which sub-population of galax-

ies dominates the cosmic ionizing emissivity. To this goal, we construct a luminosity function of $\dot{N}_{\text{ion,esc}}$, i.e., the ionizing emissivity function. This function underpins the rest of our analysis: integrating it yields the cosmic emissivity $\dot{n}_{\text{ion}}$, from which we derive the redshift evolution of the neutral fraction x_{HI} in Sect. 5.

A similar approach has already been used by [Mason et al. \(2019b\)](#) to estimate the contribution to $\dot{n}_{\text{ion}}$ from quasars, albeit with a constant and unconstrained f_{esc} . We instead apply it to the general galaxy population and measure the escaping emissivity of each galaxy directly, so that f_{esc} need not be modeled separately.

Since our spectroscopic galaxy sample is assembled from various surveys, it is extremely complicated to derive a selection function. Hence, we derive the ionizing LF based on the UV LFs at these redshifts, convolved with our measurements of the $\log(\xi_{\text{ion}}f_{\text{esc}})$ distribution. This is given by:

$$\begin{aligned} \phi(\dot{N}_{\text{ion,esc}}, z) &= \frac{dn(\dot{N}_{\text{ion,esc}}, z)}{d\dot{N}_{\text{ion,esc}}} \\ &= \int_0^\infty dM_{\text{UV}} \phi(M_{\text{UV}}, z) p(\dot{N}_{\text{ion,esc}} | M_{\text{UV}}) \end{aligned} \quad (5)$$

Here, $p(\dot{N}_{\text{ion,esc}} = L_{\text{UV}}\xi_{\text{ion}}f_{\text{esc}} | M_{\text{UV}})$ represents the distribution function of $\xi_{\text{ion}}f_{\text{esc}}$ (shown in Fig. 4) multiplied by the UV luminosity at a given M_{UV} .

We adopt this distribution at all redshifts: we cannot establish whether the mild redshift evolution of $\dot{N}_{\text{ion}}$ seen in Fig. 2 is intrinsic or a selection effect. [Simmonds et al. \(2024b\)](#) find no redshift evolution in either $\dot{N}_{\text{ion}}$ or $\xi_{\text{ion},0}$ for a mass-complete sample, hence assuming no evolution for the general population is justified.

The second key ingredient for this calculation is thus an accurate measure of the UV LF across redshift. Of particular importance are the range $z = 5$ to $z = 15$, over which the process of reionization essentially completes ([Bosman et al. 2022](#)).

For the UV LF, we adopt the measurements from the GLIMPSE survey ([Chemerynska et al. 2026](#); [Atek et al. 2026](#)) at $z > 7$. As these are based on a deep lensing-cluster field, they constrain the faint end far better than blank-field surveys, which is essential here: assessing the contribution of the faintest galaxies requires integrating down to the faintest magnitudes. We adopt a faint-end cutoff of $M_{\text{UV}} = -13$, the faintest bin reliably measured at these redshifts to date.

The GLIMPSE LFs of [Chemerynska et al. \(2026\)](#) are centered at $z = 9, 10, 11$ and 13 and include photometric candidates up to $z \sim 15$. They are fit with a double power law (DPL) rather than a Schechter function, since the latter underestimates the abundance of bright galaxies in the early Universe ([Adams et al. 2024](#); [Donnan et al. 2024](#); [Harikane et al. 2024](#); [Chemerynska et al. 2026](#)). The lower-redshift GLIMPSE LF ([Atek et al. 2026](#)) is centered at $z = 7$, covering $6 < z < 9$, and is fit with a Schechter function. At $z = 5$ and 6 we use [Bowler et al. \(2020\)](#), who adopt a DPL. For consistency across our full redshift range, we refit the $z = 7$ LF points of ([Atek et al. 2026](#)) with a DPL. The two parameterizations differ only in the brightest bins, so we expect this to have a negligible effect on our results.

The parameters of a UV LF are strongly correlated (e.g., [Bouwens et al. 2021](#); [Atek et al. 2026](#)), so treating their uncertainties as independent would overestimate the true uncertainty on the LF. To propagate the uncertainties correctly into the ionizing emissivity function, we refit every LF with a DPL using the binned data points from [Atek et al. \(2026\)](#), [Chemerynska et al. \(2026\)](#), and references therein, which gives us access to the full

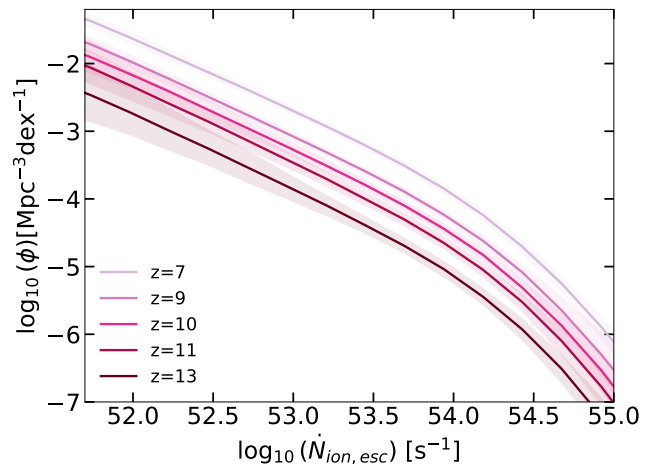


Fig. 5: Ionizing emissivity function for five redshift ranges between $7 < z < 13$, derived from the UV luminosity function, in combination with our escaping ionizing emissivity on a galaxy-per-galaxy basis. The ionizing emissivity function decreases as redshift increases, in agreement with the evolution of the UV luminosity function.

parameter chains. Because we ultimately evaluate $\dot{n}_{\text{ion}}$ at every integer redshift for a smooth interpolation, we also require the UV LF at every integer redshift. We obtain these by fitting a redshift-evolution function to each DPL parameter. The uncertainties are then calculated from the posterior distribution of the parameters. We report the DPL parameters and their uncertainties in Table A.1.

We now have all the ingredients to construct the ionizing emissivity function, which we do as follows:

1. we sample galaxies from the UV LF at a given redshift between $M_{\text{UV}} = -13$ and -23 , a range chosen to be comparable with other literature estimates ([Bouwens et al. 2015](#); [Mason et al. 2018](#));
2. we assign each M_{UV} a value of $\xi_{\text{ion}}f_{\text{esc}}$ drawn from the distribution in Fig. 4, which we apply at all magnitudes since our sample does not extend faintward of $M_{\text{UV}} = -18$;
3. we multiply the two to obtain $\dot{N}_{\text{ion,esc}}$ for each source;
4. we repeat steps (i)–(iii) 10^3 times, drawing the UV LF parameters from the refitted chains to propagate the LF uncertainties.

The resulting functions are shown in Fig. 5 for the redshifts with a GLIMPSE UV LF ([Chemerynska et al. 2026](#); [Atek et al. 2026](#)); although we derive them at every integer redshift in $5 < z < 15$. The ionizing emissivity function increases towards lower redshift, mirroring the evolution of the UV LF and indicating that more ionizing photons reach the IGM per unit time at later epochs. This is expected, since $\dot{N}_{\text{ion,esc}}$ scales with UV luminosity and we hold the distribution of $\log(\xi_{\text{ion}}f_{\text{esc}})$ fixed with redshift by construction.

We now quantify the contribution of strong leakers ($f_{\text{esc}} > 10\%$) to the ionizing emissivity function, to test whether the few sources that leak most of their ionizing photons indeed dominate it. This is not obvious a priori: although strong leakers release a large fraction of the photons they produce, they are in a minority, and the far more abundant weak leakers might collectively contribute more, despite their low f_{esc} of a few percent or less.

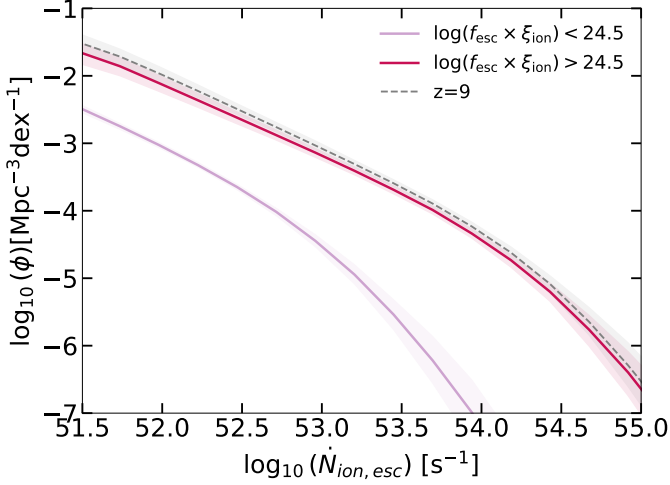


Fig. 6: Ionizing emissivity function for redshift $z=9$, split by $\log(f_{\text{esc}} \xi_{\text{ion}})$ values. The gray dashed line is the total ionizing emissivity function, as seen also in Fig. 5 while the two solid lines are the two partial emissivity functions, the red one being for $\log(f_{\text{esc}} \xi_{\text{ion}}/\text{Hz erg}^{-1}) > 24.5$ and the pink one being for $\log(f_{\text{esc}} \xi_{\text{ion}}/\text{Hz erg}^{-1}) < 24.5$.

Fig. 6 shows the same ionizing output luminosity function as in Fig. 5 for redshift $z = 9$, but split at $\log(f_{\text{esc}} \xi_{\text{ion}}/\text{Hz erg}^{-1}) = 24.5$, which corresponds to $f_{\text{esc}} \sim 10\%$ and thus separates strong leakers from weak leakers. The total emissivity is dominated by the high- f_{esc} sources above this threshold, while the contribution of the rest is negligible, especially at the highest values of $\log(\dot{N}_{\text{ion,esc}})$. We therefore confirm that, despite being far more numerous, sources with $f_{\text{esc}} < 10\%$ contribute negligibly to the rate of ionizing photons reaching the IGM. These strong leakers make up only $\sim 20\%$ of our sample.

5. Neutral gas fraction

The most commonly used probes of the global timeline of reionization are the volume-filling fraction of ionized gas, $Q_{\text{HII}}(z)$, and its counterpart, the fraction of neutral hydrogen ($x_{\text{HI}} = 1 - Q_{\text{HII}}$). To determine the fraction of neutral hydrogen and its evolution during the EoR we need to know the contribution of ionizing photons to the IGM by the entire population of galaxies, $\dot{n}_{\text{ion}}$. This quantifies the rate of ionizing photon escape from galaxies per unit time and unit volume and is usually defined as Eq. 1.

Again we do not attempt to disentangle the contributions of f_{esc} and ξ_{ion} to avoid adding unnecessary sources of uncertainty. We simply calculate $\dot{n}_{\text{ion}}$ by integrating the entire ionizing emissivity function as

$$\dot{n}_{\text{ion}} = \int \dot{N}_{\text{ion}} \phi(\dot{N}_{\text{ion}}) d\dot{N}_{\text{ion}} \quad (6)$$

which by construction of our M_{UV} sampling, is analogous to integrating down to $M_{\text{UV}} = -13$. This limit is chosen as it has been already used in the literature (Bouwens et al. 2015) to include the faintest sources while avoiding a possible turnover in the UV LF. In Fig. 7 we show the evolution of our inferred cosmic rate of ionizing photon emission as a function of redshift. At low redshift we find ourselves largely in agreement with the uncertainties of observational constraints on $\dot{n}_{\text{ion}}$ from quasar Lyman- α forests (Becker & Bolton 2013; Becker et al. 2021; Davies et al. 2024), although at $z = 6$ we are in slight tension ($\sim 1\sigma$) with

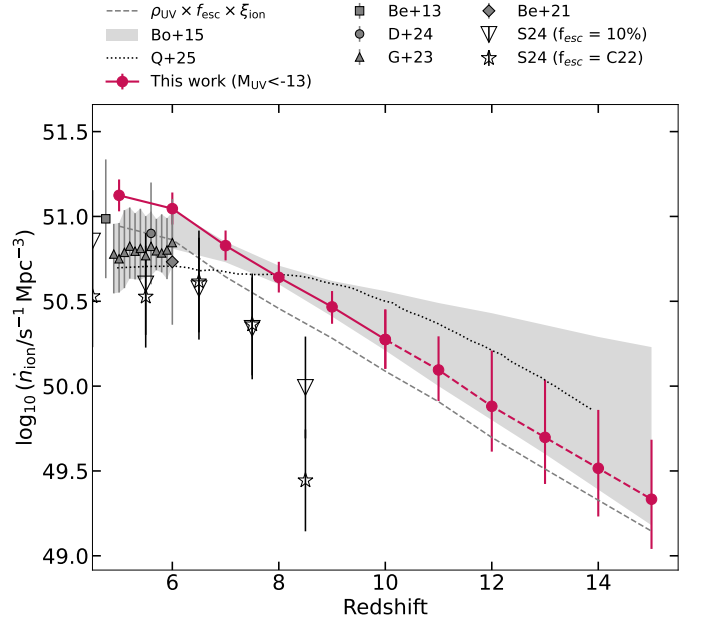


Fig. 7: Cosmic rate of ionizing photons emission per unit time and unit volume. We compare our results from other observational constraints using quasars Lyman- α forest (Becker & Bolton 2013; Becker et al. 2021; Gaikwad et al. 2023; Davies et al. 2024; Qin et al. 2025) and UV luminosity functions (Bouwens et al. 2015; Simmonds et al. 2024b), assuming a clumping factor $C_{\text{HII}} = 3$. The dashed grey line is derived by treating ρ_{UV} , f_{esc} , ξ_{ion} independently. This assumes the luminosity density from the GLIMPSE UV LFs and the means of the parameters we find in this study ($f_{\text{esc}} = 10\%$ and $\log(\xi_{\text{ion}}/\text{Hz erg}^{-1}) = 25.45$).

the observational constraints of Gaikwad et al. (2023), possibly due to their assumptions on mean free path and photo-ionization rate. At intermediate redshifts we are also in agreement with the results of Simmonds et al. (2024b). These are obtained from the Bouwens et al. (2021) UV LF and two different models for f_{esc} , one assuming a constant 10% and one assuming the relation from Chisholm et al. (2022). Above $z > 6$ we are entirely in agreement with the estimates from Bouwens et al. (2015), who assume a clumping factor $C_{\text{HII}} = 3$, as we do. At $z > 8$ we find a broad agreement with the models based on XQR-30 data from Qin et al. (2025). In the same figure we also show the evolution of $\dot{n}_{\text{ion}}$ we would get by treating the three factors independently (i.e. using Eq. 1). For this calculation we use the mean parameters we find from our SED fitting ($f_{\text{esc}} = 10\%$ and $\log(\xi_{\text{ion}}/\text{Hz erg}^{-1}) = 25.45$). We find these results to be more in agreement with those from Simmonds et al. (2024b) and the Lyman- α forest (Becker & Bolton 2013; Becker et al. 2021; Davies et al. 2024), while not being consistent with results from Bouwens et al. (2015) and Qin et al. (2025). This shows that it is necessary to account for the correlation of the parameters in this calculations as it significantly affects the inferred cosmic rate of ionizing photons emission.

From $\dot{n}_{\text{ion}}$ it is relatively straightforward to compute the volume-filling fraction of ionized gas by solving the differential equation

$$\dot{Q}_{\text{HII}} = \frac{\dot{n}_{\text{ion}}}{\langle n_{\text{H}} \rangle} - \frac{Q_{\text{HII}}}{t_{\text{rec}}} \quad (7)$$

where $\langle n_H \rangle$ is the average hydrogen number density and t_{rec} is the recombination time. t_{rec} is defined as

$$t_{\text{rec}} = [C_{\text{HII}}\alpha_B(T)(1 + Y_p/4X_p)\langle n_H \rangle(1 + z)^3]^{-1}. \quad (8)$$

For this calculation we assume a recombination coefficient for a temperature of $T = 10^4$ K of $\alpha_B(T) = 2.6 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ from [Osterbrock & Ferland \(2006\)](#), primordial abundances of hydrogen and helium such that $1 + Y_p/4X_p = 1.08$ and a clumping factor $C_{\text{HII}} = 3$, motivated by results from simulation ([Finlator et al. 2012](#); [Shull et al. 2012](#)). It should be noted that other studies have shown that the clumping factor may be higher than $C_{\text{HII}} = 3$. [Austin et al. \(2025\)](#) have shown that, to reionize the Universe by $z = 6$, the clumping factor may be 6.2 or even as high as 8.8, depending on the model used for f_{esc} . It has also been shown from models and simulations that C_{HII} is likely to evolve, increasing from ~ 1 to ~ 8 as reionization proceeds ([Pawlik et al. 2015](#); [Qin et al. 2025](#)). An increase of clumping factor would affect our x_{HI} calculations by postponing the end of reionization by $\Delta z = 0.4$ for $C_{\text{HII}} = 6$.

Solving Eq. 7 we find the evolution of Q_{HII} and therefore of x_{HI} with redshift. Our results are presented in Fig. 8. Here we compare them with other literature values for a variety of probes of the ionization state of the IGM such as the Lyman- α emitters clustering ([Sobacchi & Mesinger 2015](#); [Ouchi et al. 2018](#)), Lyman-break galaxies damping wing ([Hsiao et al. 2024](#); [Umeda et al. 2024](#); [Mason et al. 2026](#)), the Lyman- α equivalent width evolution ([Mason et al. 2018, 2019a](#); [Hoag et al. 2019](#); [Whitler et al. 2020](#); [Bolan et al. 2022](#); [Bruton et al. 2023](#); [Morishita et al. 2023](#); [Nakane et al. 2024](#); [Tang et al. 2024](#); [Kageura et al. 2025](#)), Lyman- α luminosity functions ([Ouchi et al. 2010](#); [Konno et al. 2014, 2018](#); [Zheng et al. 2017](#); [Inoue et al. 2018](#); [Goto et al. 2021](#); [Morales et al. 2021](#); [Ning et al. 2022](#)), Lyman- α forest dark pixel fraction ([Jin et al. 2023](#)), quasar damping wings ([Schroeder et al. 2013](#); [Davies et al. 2018](#); [Wang et al. 2020](#); [Greig et al. 2019, 2024](#)) and the Lyman- α forest ([Qin et al. 2025](#)). Overall we find our results to be broadly consistent with these estimates. We find an end to reionization at $z \sim 5.8$, also in agreement with recent estimates from quasar spectra such as [Bosman et al. \(2022\)](#), who find an end to reionization at $z \sim 5.3$ but $x_{\text{HI}} \sim 6 \times 10^{-5}$ at $z \sim 5.8$.

Surprisingly, we find that the largest source of uncertainty in the calculation of the neutral hydrogen fraction in this framework is not the uncertainty on f_{esc} and ξ_{ion} , but that on the UV luminosity density; in particular, the uncertainty on the faint end slope of the LF. In Fig. 8 we show three shaded regions, each representing a different uncertainty. The smallest shaded area represents the uncertainty which does not consider any uncertainty in the UV LF, using only the best fit parameters. The second shaded area shows the uncertainty in the estimation of x_{HI} when we also consider the uncertainty in the GLIMPSE LF. Lastly, we use the same method detailed in the previous section with the LF parameters from [Donnan et al. \(2024\)](#). These LFs are much less constrained at the faint end. This is due to the lack of very faint sources, as their data does not include very deep and lensed fields such as the GLIMPSE field. In this case, the uncertainty on x_{HI} significantly increases, showing that constraining the faint end of the LF at high redshift is necessary to determine the timeline and evolution of reionization. The values of x_{HI} with the associated uncertainties for both GLIMPSE and [Donnan et al. \(2024\)](#) are tabulated for all integer redshifts in Tab. B.1.

We also compare the redshift evolution of the neutral fraction to the standard estimate that only uses average values rather than the actual distribution functions, i.e., $\dot{n}_{\text{ion}} = \rho_{\text{UV}} f_{\text{esc}} \xi_{\text{ion}}$. For

the UV luminosity density we again integrate the GLIMPSE LF, while for other quantities we use the mean values for our samples of $f_{\text{esc}} = 0.1$ and $\log(\xi_{\text{ion}}/\text{Hz erg}^{-1}) = 25.45$. As already discussed, treating these parameters independently significantly affects the reionization calculations. This is also seen here, where we find a delayed progress of reionization ending at $z \sim 5.4$. As all the quantities in the calculation are derived from the same information in the spectrum, the line fluxes and the UV slope, they must not be treated independently, to avoid underestimating the ionizing $\dot{n}_{\text{ion}}$ and to accurately describe reionization.

It is also possible to compare this reionization history with cosmological constraints by measuring the optical depth of electron scattering as measured from the CMB τ_{CMB} . It is defined as ([Robertson 2022](#)):

$$\tau(z) = c \langle n_H \rangle \sigma_T \int_0^z f_e Q_{\text{HII}}(z') H^{-1}(z') (1 + z')^2 dz' \quad (9)$$

where c is the speed of light, $\langle n_H \rangle$ is the comoving average hydrogen number density, σ_T is the Thomson cross-section, $H(z)$ is the redshift dependent Hubble parameter and f_e the number of free electrons per hydrogen atom, which depends on the ionization state of hydrogen and helium. We find a value of τ_{CMB} consistent with results from [Planck Collaboration et al. \(2020\)](#) but much lower than a recent value suggested by ([Sailer et al. 2026](#)), based on DESI data, which would solve the neutrino mass tension. This value of τ is itself in a $\sim 2\sigma$ tension with other estimates ([Cain et al. 2025b](#)).

5.1. Contributions to the neutral gas fraction

One of the biggest open questions in the study of the EoR is understanding which kind of sources contribute most to the reionization process. We try to answer this question by quantifying the contribution of strong leakers to the evolution of the neutral hydrogen fraction over the history of the Universe. This will tell us whether it is the few highly leaking sources or the many weakly leaking sources that drive the bulk of the reionization process. Similarly to what was already done in Fig. 6 we split our sampling between sources with $\log(\xi_{\text{ion}} f_{\text{esc}}/\text{Hz erg}^{-1}) < 24.5$ and those above that value and calculate what the evolution of the neutral hydrogen fraction for each sample (see Fig 9). We find that reionization is almost entirely driven by the few sources with a high escape fraction. These sources alone can ionize the universe in agreement with the observational constraints. The majority of sources, those with $f_{\text{esc}} \lesssim 0.1$, do not have a significant contribution to the ionization of neutral gas in the universe. This shows that it is in fact the few sources with very high escape fraction, $\sim 20\%$ of the total sample, that perform the bulk of reionization, producing $\sim 87\%$ of all ionizing photons in the EoR, instead of the majority of sources with low escape fraction. This is broadly in agreement with recent results from the PRISMS survey, who also find that a minority of sources with very high f_{esc} should be responsible for the majority of the reionization process ([Marques-Chaves et al. 2026](#)). It should be noted that high f_{esc} is likely a temporal event. Simulations in fact show that f_{esc} can drastically change in galaxies as a function of time, usually following star formation events ([Trebitsch et al. 2017](#)). It would therefore not be the same 20% of sources driving all of the reionization process but nonetheless, at any given time, we expect that only a minority of sources strongly contribute to this process.

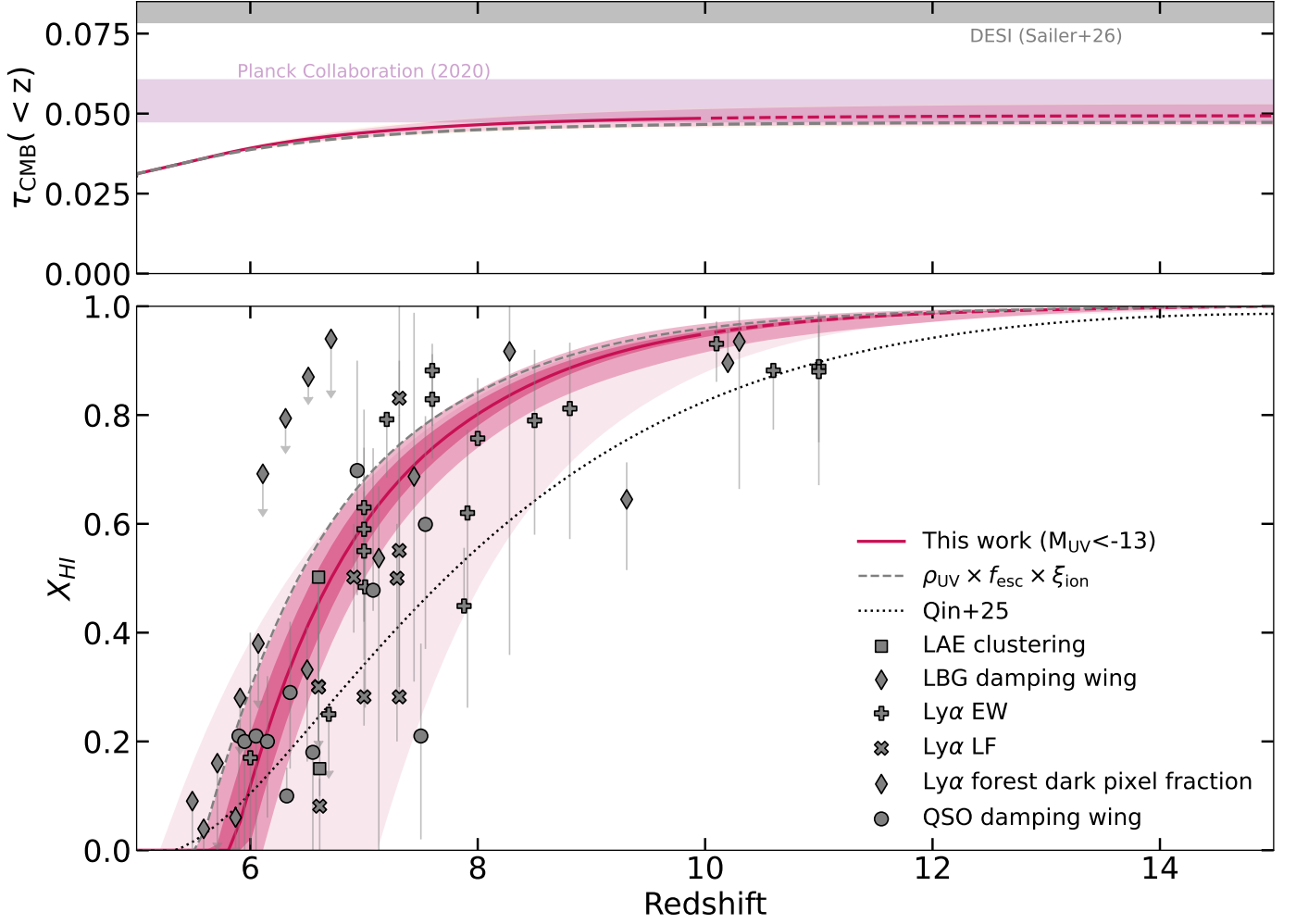


Fig. 8: **Top panel:** Evolution of the optical depth of electron scattering as a function of redshift. Our results agree with constraints from Planck Collaboration et al. (2020), in the pink shaded area, but are much lower than recent values hypothesized by Sailer et al. (2026), based on DESI data, to solve the neutrino mass tension, shown in grey. **Bottom panel:** Neutral hydrogen fraction as a function of redshift, compared to literature points. Our results, down to an integration limit of $M_{\text{UV}} = -13$ are shown with the red line, where the dashed section represents the redshift range in which we have extrapolated our results. The lightest shaded area represents the uncertainty when using the UV LF from Donnan et al. (2024). The middle shaded area represents the uncertainty using the GLIMPSE LFs (Chemerynska et al. 2026; Atek et al. 2026). The smallest and darkest shaded area represents the uncertainty if we assume no uncertainty on the UV LF, indicating that this is in fact the largest source of uncertainty in this calculation. The gray dashed line shows the evolution estimated with just average values, i.e. ρ_{UV} from the UV LF, $f_{\text{esc}} = 10\%$ and $\xi_{\text{ion}} = 25.45$, the means of the parameters as found from the SED fitting. The gray points show literature estimates of x_{HI} from various probes (see main text for references). The dotted black line shows the neutral hydrogen fraction derived in (Qin et al. 2025).

5.2. Do faint or bright galaxies drive reionization?

It is still unclear at what magnitude range galaxies contribute the most to the reionization, whether it is faint (Atek et al. 2024) or bright galaxies (Naidu et al. 2020). Recent work targeting the faintest galaxies finds that those do not have the highest escape fractions and that the largest contribution of ionizing photons comes from galaxies with $M_{\text{UV}} \sim -16$ (Jecmen et al. 2026). In Fig. 10 we investigate which sources contribute the most to the reionization process. First, in the left panel, we show the cumulative distribution of $\dot{n}_{\text{ion}}$ as a function of M_{UV} for nine integer redshifts between $5 < z < 13$. Overall we see that the amount of ionizing photons emitted into the IGM increases as redshift decreases, as we already saw in Fig. 5. We also see that in general very bright galaxies do not contribute strongly to the total amount of ionizing photons in a unit volume. Moreover, we find

that the cumulative distribution flattens at different magnitudes for the different redshift. We note that as we have assumed no dependence of $\xi_{\text{ion}} f_{\text{esc}}$ on M_{UV} these results would be the same for non ionizing UV radiation. This indicates that at higher redshifts faint galaxies have a larger contribution than at the end of reionization, where we find the distribution already flattening at $M_{\text{UV}} \sim -18$. In Fig. 10, we also show the normalized fractional contribution of sources in bins of $0.5 M_{\text{UV}}$ to the total $\dot{n}_{\text{ion}}$. Here we can again see that the fractional contribution varies with redshift, with bright sources being more important at lower ($z \leq 6$) redshifts. Indeed, for the case of $z = 5$ we find that the largest contribution to the total $\dot{n}_{\text{ion}}$ comes from galaxies with $M_{\text{UV}} \sim -20$. This is also true for $z = 6$ although the contribution of fainter sources is more significant than at the lowest redshift. Above $z > 6$ we find that bright galaxies do not strongly con-

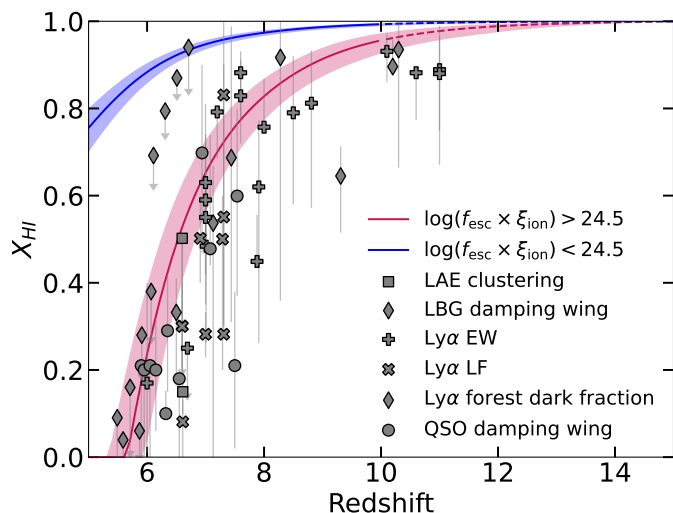


Fig. 9: Neutral hydrogen fraction as a function of redshift split at $\log(f_{\text{esc}}\xi_{\text{ion}}/\text{Hz erg}^{-1}) = 24.5$. The red line only includes the sources with $\log(f_{\text{esc}}\xi_{\text{ion}})$ above this value while the blue line only includes sources below this value. This shows that strong leakers ($f_{\text{esc}} \gtrsim 0.1$) alone do the majority of the reionization process, while the contribution of the remaining sources is not significant.

tribute to the total $\dot{n}_{\text{ion}}$ while fainter sources, with $M_{\text{UV}} > -19$ equally contribute. This trend is consistent with the constant $f_{\text{esc}} = 14\%$ from Jecmen et al. (2026). While their fiducial distribution shows a peak at $M_{\text{UV}} = -16$, which ours doesn't, they also find that sources with M_{UV} between -18 and -14 drive reionization, which is consistent with our results, within the limits of extrapolation. It is also in agreement with recent results from the THESAN-ZOOM simulation, which show that brighter galaxies contribute more at lower redshifts (Summerfield et al. 2026).

Overall, this tells us that faint galaxies contribute more at high redshifts while at lower redshifts their contribution decreases in favor of brighter galaxies. It is important to note that we do not assume any magnitude evolution for $\log(\xi_{\text{ion}}f_{\text{esc}})$, effectively assuming that faint galaxies are as efficient in emitting ionizing photons as bright galaxies. The reason why they dominate the contribution of ionizing photons to the IGM is that, although they individually produce less ionizing photons, they are much more numerous, especially at high redshift, where very bright sources are increasingly rarer. This trend is therefore highly dependent on the faint end slope of the luminosity function. Indeed, if we compare our results to those based on [OIII]+H β luminosity function we find that they are inconsistent. Both Korber et al. (2026) and Meyer et al. (2024) find that, when deriving $\dot{n}_{\text{ion}}$ from the [OIII]+H β luminosity function, bright galaxies contribute more at higher redshifts. This discrepancy is due to the faint end of the [OIII]+H β and UV luminosity functions diverging at high redshift, with the UV LF having a steeper slope, increasing the contribution of faint sources in our model. This discrepancy reinforces the need for better constraints on the faint end slope of UV LFs at high redshift.

6. Caveats

Here we would like to discuss the main assumptions and caveats of this work.

First of all we would like to once again note, as already discussed in Giovanazzo et al. (2026), the degeneracy between star formation history and escape fraction, as this would affect our estimates of $\dot{N}_{\text{ion,esc}}$. However, we had already found that this uncertainty is less important for very high f_{esc} sources, which often have spectra that cannot be reproduced with $f_{\text{esc}} = 0$. Therefore, as we have found that the sources with $f_{\text{esc}} > 10\%$ are the main drivers of reionization, we do not think that this degeneracy will have a strong impact on our results. Moreover, there is a category of sources, remnant leakers (Katz et al. 2023), for which this degeneracy is not important, as they are quenched sources with nonetheless high f_{esc} . These sources would have still young stellar populations but little ISM, making them efficient leakers.

We would also like to note that since our method to calculate $\dot{N}_{\text{ion,esc}}$ relies on SED fitting, it includes assumptions on the stellar populations, initial mass functions (IMFs) and dust attenuation laws. We assume a BPASS v2.2.1 stellar population (Stanway & Eldridge 2018) with the default broken power law IMF, so if these assumptions were to not be accurate it could affect our results. In this context, our quantitative conclusions should be interpreted within our modeling framework, although we do not believe that our qualitative conclusions on the sources of reionization would be affected.

Possibly the biggest caveat of the analysis presented in this work is the assumption of a lack of magnitude evolution for $\log(\xi_{\text{ion}}f_{\text{esc}})$. This assumption allows us to expand the analysis to magnitudes not covered by our initial sample and investigate the impact of the faintest galaxies to reionization. However, just because we find no evidence of a trend between $\log(\xi_{\text{ion}}f_{\text{esc}})$ and M_{UV} in our sample it does not necessarily mean that this will be true at all magnitudes. Overcoming this would require an analysis of the ionizing output of sources covering the fainter end of the magnitude distribution that we are lacking here. Moreover, due to our choice to use only spectra with a confident redshift estimate, we might be biasing our sample to only strong line emitters, which would also bias our $\log(\xi_{\text{ion}}f_{\text{esc}})$ distribution.

Lastly, we would like to mention the issue of cosmic variance. As we have seen that the estimate of the neutral hydrogen fraction is very sensitive to the UV luminosity function uncertainty, especially in the faint end slope, we have chosen to use the luminosity function from GLIMPSE (Chemerynska et al. 2026; Atek et al. 2026). However, GLIMPSE is only one NIR-Cam pointing of a lensed cluster, which significantly reduces the effective volume. It could therefore be highly affected by cosmic variance, although this effect is less problematic for very faint sources. Alleviating this issue would require many more extremely deep field observations with JWST or the upcoming *Extremely Large Telescope* (ELT).

7. Conclusion

In this work we leverage JWST/NIRSpec spectra from a variety of public observations in the redshift range $5 < z < 10$, retrieved from the DJA, to compile a sample of 1428 sources. Using the SED fits already performed in Giovanazzo et al. (2026) we investigate the ionizing photon production of galaxies in the Epoch of Reionization. We estimate the ionizing photon output of individual galaxies, accounting for their intrinsic ionizing photon production and their escape fraction. Starting from the GLIMPSE UV luminosity functions (Chemerynska et al. 2026; Atek et al. 2026), as a way to account for the incompleteness of our sample, we also compute the ionizing emissivity function. This quantifies the amount of sources emitting a given rate of ionizing photons in the IGM per unit time per unit volume. We also determine the

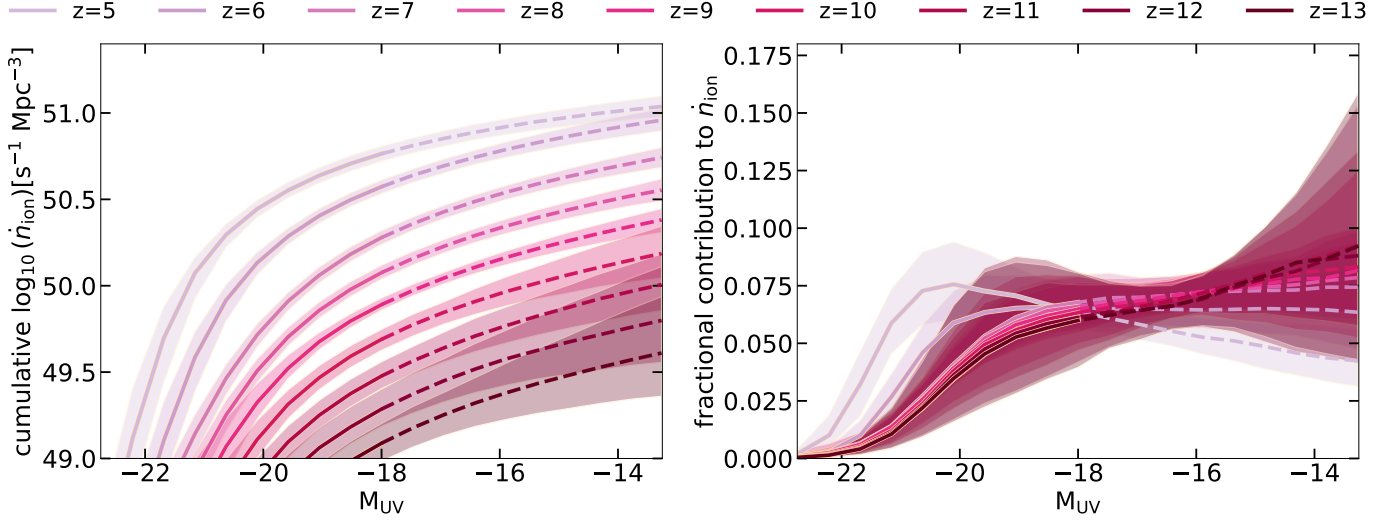


Fig. 10: **Left panel:** cumulative distribution of $\dot{n}_{\text{ion}}$ as a function of M_{UV} for nine integer redshift in the range $5 < z < 13$. For all the lines, the dashed section represents the magnitude range in which we have extrapolated our results, as our initial sample only includes sources with $M_{\text{UV}} < -18$. Overall, the amount of ionizing photons emitted per unit time and unit volume increases at lower redshift. Also, the contribution of fainter sources at lower redshift becomes less important compared to bright sources. **Right panel:** fractional contribution to the total $\dot{n}_{\text{ion}}$ at each magnitude for the same redshifts, in bins of $\Delta M_{\text{UV}} = 0.5$. The dashed sections once again represent the range in which our results are extrapolated. While at high redshift, faint sources ($M_{\text{UV}} \gtrsim -18$) contribute the most to the total $\dot{n}_{\text{ion}}$, at low redshift bright sources contribute the most.

importance that high f_{esc} sources ($f_{\text{esc}} > 0.1$) have for the reionization process and whether, although not very numerous, they drive it. Our main findings are summarized below.

- From our SED fits we calculate the amount of ionizing photon produced by each galaxy, as well as the amount of ionizing photons emitted into the IGM, when the escape fraction is taken into account. We find that the intrinsic production shows a slight evolution towards higher values at higher redshifts. This is consistent with the results of [Simmonds et al. \(2024b\)](#), who also use a spectroscopic only sample, possibly indicating that this trend is due to the incompleteness and selection function of the sample. We find a similar trend for the escaped rate of ionizing photons, which also increases as a function of redshift, although at a different rate for $f_{\text{esc}} < 0.1$ and $f_{\text{esc}} > 0.1$. Indeed we find a stronger evolution, more consistent with the intrinsic, for strong leakers, while weak leakers show an almost flat trend.
- We find that in general the sources with higher f_{esc} are in fact those that emit the most ionizing photons into the IGM, which is an expected although not obvious result. At a constant UV magnitude we find a cloud of low f_{esc} sources about ~ 1.5 dex below the leakers. This translates into the presence of two clouds, one of strong leakers and one of weak leakers also in $\log_{10}(\xi_{\text{ion}} f_{\text{esc}})$.
- The ionizing emissivity function decreases with increasing redshift, following the behavior of the UV luminosity function. We study the behavior of this function when splitting the sample at $\log_{10}(\xi_{\text{ion}} f_{\text{esc}}) = 24.5$, roughly corresponding to $f_{\text{esc}} \sim 0.1$ split and the mean of the sample. In this case we find that the main contribution to the rate of ionizing photons emitted into the IGM per unit volume comes almost entirely from sources above these values. Only at low $\dot{N}_{\text{ion}}$ values low f_{esc} sources start contributing.
- Based on the ionizing emissivity function we estimate the evolution of the fraction of neutral hydrogen with redshift.

We find its evolution to be consistent with literature estimates and that the uncertainty on the UV luminosity function is the lead cause of uncertainties in our estimates, especially the uncertainty in the faint end slope. We compare this evolution to that found when assuming $f_{\text{esc}} = 0.1$ and $\log(\xi_{\text{ion}} / \text{Hz erg}^{-1}) = 25.45$ as motivated from our results. We find them to be in broad agreement with the latter being faster. We also estimate τ_{CMB} and find it in agreement with the results from [Planck Collaboration et al. \(2020\)](#).

- We determine the contribution to the ionized hydrogen fraction by strong leakers by performing again the same analysis over two samples split at $\log(\xi_{\text{ion}} f_{\text{esc}} / \text{Hz erg}^{-1}) = 24.5$. We find that the strong leakers ($f_{\text{esc}} > 0.1$) alone are almost able to match the reionization history that we find for all the sources. This shows that the few strongly leaking sources, $\sim 20\%$ of the total sources in the universe, are able to entirely drive the reionization process, producing $\sim 87\%$ of all ionizing photons that reach the IGM. This indicates that it is necessary to identify highly leaking galaxies to find the drivers of reionization.
- Lastly, we study the contribution to the total $\dot{n}_{\text{ion}}$ from sources of different magnitudes. We find that at high redshift sources with $M_{\text{UV}} > -18$ all seem to contribute equally, while at low redshift bright sources ($M_{\text{UV}} \sim -20$) contribute most.

With this work we have presented a framework to calculate the ionizing photon emission of individual galaxies, as well as the rate of ionizing photons being emitted in a unit volume. Our results are consistent with observational constraints and show that 20% of galaxies with high f_{esc} are almost entirely responsible for reionization, rather than the majority of low ionizing output galaxies. The relative contributions of faint and bright galaxies vary with redshift, with bright sources dominating at the end of reionization and faint sources at its onset. Pinning down the drivers of reionization will ultimately require pushing f_{esc} measurements to the faint, abundant galaxies that our spectroscopic

sample cannot yet reach, a regime that deep lensed-field spectroscopy with JWST, and eventually the ELT, is poised to study in the future.

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Appendix A: UV LF parameters

In Table A.1 we compile the parameters and related uncertainties of the double power law parameters used in the analysis.

Table A.1: DPL parameters and uncertainties

Redshift	M^*	$\log_{10}(\phi^*)$	α	β
5	$-21.52^{+0.39}_{-0.37}$	$-3.52^{+0.25}_{-0.23}$	$-1.90^{+0.08}_{-0.08}$	$-5.37^{+0.32}_{-0.38}$
6	$-21.03^{+0.39}_{-0.37}$	$-3.52^{+0.25}_{-0.23}$	$-1.99^{+0.08}_{-0.08}$	$-4.92^{+0.32}_{-0.38}$
7	$-20.39^{+0.39}_{-0.37}$	$-3.48^{+0.25}_{-0.23}$	$-2.01^{+0.08}_{-0.08}$	$-3.89^{+0.32}_{-0.38}$
8	$-20.42^{+0.39}_{-0.37}$	$-3.70^{+0.25}_{-0.23}$	$-2.02^{+0.08}_{-0.08}$	$-3.96^{+0.32}_{-0.38}$
9	$-20.45^{+0.25}_{-0.28}$	$-3.93^{+0.18}_{-0.23}$	$-2.04^{+0.08}_{-0.07}$	$-4.03^{+0.33}_{-0.49}$
10	$-20.49^{+0.31}_{-0.37}$	$-4.15^{+0.23}_{-0.27}$	$-2.06^{+0.15}_{-0.14}$	$-4.10^{+0.48}_{-0.48}$
11	$-20.52^{+0.39}_{-0.30}$	$-4.38^{+0.29}_{-0.29}$	$-2.07^{+0.13}_{-0.14}$	$-4.17^{+0.46}_{-0.48}$
12	$-20.55^{+0.40}_{-0.42}$	$-4.61^{+0.30}_{-0.31}$	$-2.09^{+0.22}_{-0.21}$	$-4.24^{+0.47}_{-0.48}$
13	$-20.58^{+0.40}_{-0.42}$	$-4.83^{+0.30}_{-0.31}$	$-2.10^{+0.22}_{-0.21}$	$-4.31^{+0.47}_{-0.48}$
14	$-20.61^{+0.40}_{-0.42}$	$-5.06^{+0.30}_{-0.31}$	$-2.12^{+0.22}_{-0.21}$	$-4.38^{+0.47}_{-0.48}$
15	$-20.65^{+0.40}_{-0.42}$	$-5.29^{+0.30}_{-0.31}$	$-2.14^{+0.22}_{-0.21}$	$-4.45^{+0.47}_{-0.48}$

Appendix B: Neutral gas fraction

Table B.1: x_{HI} and $\log(\dot{n}_{\text{ion}})$ evaluated at integer redshifts with uncertainties

z	$\log(\dot{n}_{\text{ion}}/\text{s}^{-1}\text{Mpc}^{-3})$	x_{HI}	GLIMPSE	D24
15	$49.3^{+0.35}_{-0.29}$	1.00	± 0.00	± 0.00
14	$49.41^{+0.34}_{-0.28}$	0.99	± 0.01	± 0.01
13	$49.70^{+0.33}_{-0.27}$	0.99	± 0.01	$+0.01$
12	$49.88^{+0.32}_{-0.27}$	0.98	$+0.01$	-0.02
11	$50.10^{+0.20}_{-0.18}$	0.97	$+0.02$	$+0.01$
10	$50.27^{+0.17}_{-0.17}$	0.94	$+0.03$	-0.04
9	$50.47^{+0.09}_{-0.10}$	0.88	$+0.04$	$+0.02$
8	$50.64^{+0.09}_{-0.09}$	0.76	$+0.06$	$+0.04$
7	$50.83^{+0.09}_{-0.09}$	0.51	$+0.10$	$+0.08$
6	$51.05^{+0.09}_{-0.09}$	0.00	$+0.14$	$+0.26$
5	$51.12^{+0.09}_{-0.09}$	0.00	-0.00	-0.00
			± 0	± 0.00

Notes. The evolution of x_{HI} and $\log(\dot{n}_{\text{ion}})$ evaluated at integer redshift. For x_{HI} we also show the uncertainties both for the case of the GLIMPSE LF and for the case of the [Donnan et al. \(2024\)](#) (D24) LF, as also seen in Fig. 8.